LESLEY VAN HOEK

Software engineer, researcher, and artist in creative technology

May 21, 1995 | Rotterdam, The Netherlands | lesleyvanhoek.nl | lesleyvanhoek@hotmail.com | lshoek

EMPLOYMENT

Senior Software Engineer
Studio DRIFT

Apr. 2024 – present | Amsterdam, The Netherlands

- Build high-performance software for permanent large-scale real-time audiovisual and motor-controlled installation artworks with C++/NAP.
- Solve unique complex problems w/ custom algorithms/solutions i.e. custom render pipelines, 3D fluid simulation, EtherCAT MainDevice etc.

Software Engineer
NAP Labs + NAP Framework

Jan. 2020 – Apr. 2024 | Amsterdam, The Netherlands

- NAP is an open-source real-time control & visualization platform used to build large audiovisual installations & control high-precision robotics.
- Made large contributions to NAP (mostly the Vulkan render engine) & built software for many demanding professional art installations.

Unity Developer
TWNKLS Augmented Reality

Dec. 2017 – Sept. 2018 | Rotterdam, The Netherlands

- Developed AR applications for mobile devices, among which 'Augmented Blueware' for Royal Delft.

Graphics Developer
NEP Graphics

Dec. 2016 – Sept. 2017 | Hilversum, The Netherlands

- Contributed to graphics systems deployed at various customers under which the Dutch national broadcasting station (NOS).

Software Development intern
RE'FLEKT GmbH

Feb. 2016 – Jul. 2016 | Munich, Germany

- Case study: Enhancing visual realism of virtual objects in AR envs.

EDUCATION

MSc Creative Intelligence & Technology
Leiden University

Sept. 2018 – Sept. 2020 | Leiden, The Netherlands

- Wrote my master's thesis "Evolving Virtual Embodied Agents using External Artifact Evaluations."
- Completed the courses: Deep Learning & Neural Networks; Evolutionary Algorithms; Human Computer Interaction; Non-Human Cognition; Scientific Narration & Visualization; Sound, Space & Interaction etc.
- Worked as a teaching assistant for the courses "Introduction to Video Game Making" and "New Media & New Technologies."

SOFTWARE

Programming languages

C++ | C | Python | C#

Tools & frameworks

NAP | Vulkan | OpenGL/GLSL | OpenCV | Tensorflow | Pytorch | Bullet Physics | openFrameworks | Unity | Arduino

Version control

Git

LANGUAGES

Dutch
Native language

English
C2 full professional proficiency (certified, University of Cambridge)

German
B1 limited working proficiency (certified, Stuttgart Media University)

OTHER

Driver's license
Category B

PORTFOLIO

lesleyvanhoek.nl
Please refer to my homepage for a more extensive overview of my technical, scientific and artistic work.

BSc Computer Science

Avans University of Applied Sciences

Sept. 2012 – Aug. 2016 Breda, The Netherlands

- Wrote a case study report "Enhancing Visual Realism of Virtual Objects in Mobile Augmented Reality Applications using Unity3D."
- Completed the courses: Algorithms; 3D Computer Graphics; Computer Vision; Mixed Reality; Databases, Design Patterns; Network technology; Digital Signal Processing; User Experience & Interaction etc.

Minor Games Technology

Stuttgart Media University

Sept. 2015 – Feb. 2016 Stuttgart, Germany

- Completed the courses: Game Engine Programming; Computer Graphics; Programming Massively Parallel Processors; Gameplay Programming; Practical & Theoretical Game Development etc.

ACCOMPLISHMENTS

Publications

- van Hoek, L., Saunders, R. & de Kleijn, R. (Forthcoming).“Evolving Virtual Embodied Agents using External Artifact Evaluations”. Springer Communications in Computer and Information Science.

Awards

- Best Video Award. BNAIC/BeneLearn 2020 (Benelux Conference on Artificial Intelligence and Machine Learning)

EXHIBITIONS

Sonaural
And so on, and so forth; WORM, Rotterdam (2020)

Certainty Expo
And so on, and so forth; _V2, Rotterdam (2019)

Parallels Expo
Artefacts of Verisimilitudes; Peron E, Utrecht (2019)